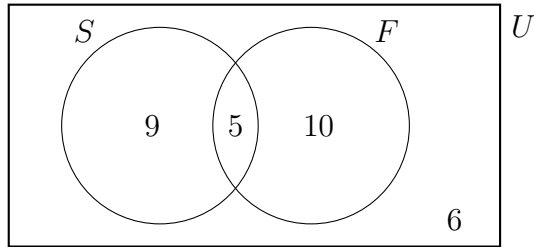


4.18 Homework: Probability Review

This is practice for Friday's quiz. Show all working.

1. In a class of 30 students, some study Spanish (S) and some study French (F). Five students study both languages.



(a) Write down the number of students who study Spanish. [1]

(b) One student is chosen at random. Find the probability that

(i) the student does not study Spanish; [1]

(ii) the student studies Spanish or French, but not both. [2]

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2. Events C and D are mutually exclusive with $P(C) = 0.35$ and $P(D) = 0.25$.

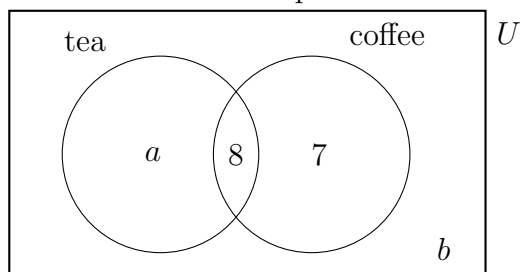
(a) Write down $P(C \cap D)$. [1]

(b) Find $P(C \cup D)$. [2]

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3. In a group of 40 people, 22 like tea and 15 like coffee. Eight people like both, as shown in the Venn diagram. The values a and b represent numbers of people.



- (a) Find the value of a . [1]
- (b) Find the value of b . [1]
- (c) A person is chosen at random. Find the probability that the person likes coffee but not tea. [1]
- (d) Find the probability that a person likes tea given that they like coffee. [2]

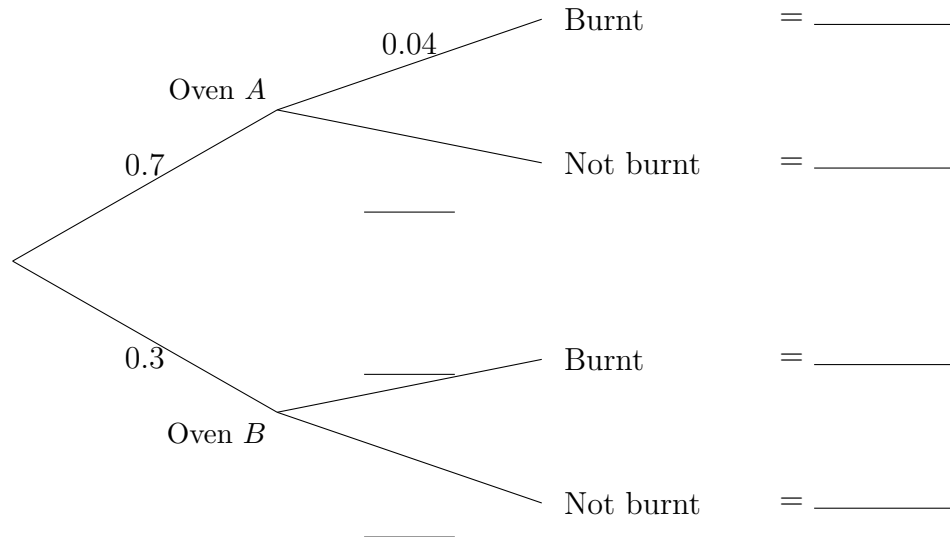
4. The following table shows the travel method and punctuality of 20 students.

	Punctual (P)	Sometimes late (T)	Total
Walk	3	7	10
Bus	2	8	10
Total	5	15	20

- (a) Find the probability that a randomly chosen student walks to school and is punctual. [1]
- (b) Find $P(\text{punctual} \mid \text{bus})$. [2]
- (c) Determine whether “walks to school” and “is punctual” are independent events. Show your working and justify your answer. [3]

5. A bakery makes bread using two ovens. Oven A bakes 70% of the loaves and oven B bakes the remaining 30%. The probability that a loaf from oven A is burnt is 0.04. The probability that a loaf from oven B is burnt is 0.10.

(a) Complete the tree diagram below, showing all probabilities. [3]



(b) A loaf is selected at random. Find the probability that the loaf is

(i) burnt and from oven A ; [2]

(ii) burnt. [2]

(c) Given that a loaf is burnt, find the probability that it came from oven A . [3]

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6. A bag contains 5 red marbles and 7 blue marbles. Two marbles are drawn at random, one after the other, without replacement.
- (a) Find the probability that both marbles are red. [2]
- (b) Find the probability that both marbles are the same colour. [3]
- (c) Find the probability that both marbles are red given that at least one marble is red. [3]